

Circuit Designs

Laboratory-Scale , Three-Phase Grid-Forming Inverter for Islanded Microgrid Applications

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Table of Contents

1	Controlling Circuit	4
1.1	Schematic Diagram	4
1.2	PCB design	5
1.3	Final PCB	6
1.4	Operation.....	7
1.5	Components	10
2	Inverter Circuit	11
2.1	Schematic Diagram	11
2.2	PCB design	12
2.3	Final PCB	12
2.4	Operation.....	13
2.5	Components	14
3	LCL filter + Voltage and Current measure circuit	15
3.1	Schematic Diagram	15
3.2	PCB design	16
3.3	Final PCB	16
3.4	Operation.....	17
3.5	Components	18

List of Figures

Figure 1: Schematic diagram of Controlling circuit	4
Figure 2: Controller circuit : PCB : Layer 1	5
Figure 3: Controller circuit : PCB : Layer 2	5
Figure 4: Hardware - Controller circuit	6
Figure 5: Gate driver circuit	7
Figure 6: MP1584EN DC-DC buck converter	7
Figure 7: UA7805CKCS.....	7
Figure 8: Schematic diagram of Inverter circuit.....	11
Figure 9: PCB design of Inverter circuit.....	12
Figure 10: Hardware circuit - Inverter circuit.....	12
Figure 11: Schematic diagram : LCL + voltage, current measuring circuit	15
Figure 12: PCB design : LCL filter + voltage, current measurement circuit.....	16
Figure 13: Hardware circuit : LCL + voltage , current measurement circuit	16

List of Tables

Table 1: Components of Controller Circuit	10
Table 2: Components of Inverter Circuit	14
Table 3: Components of LCL + Voltage and current measure circuit	18

1.1 Schematic Diagram

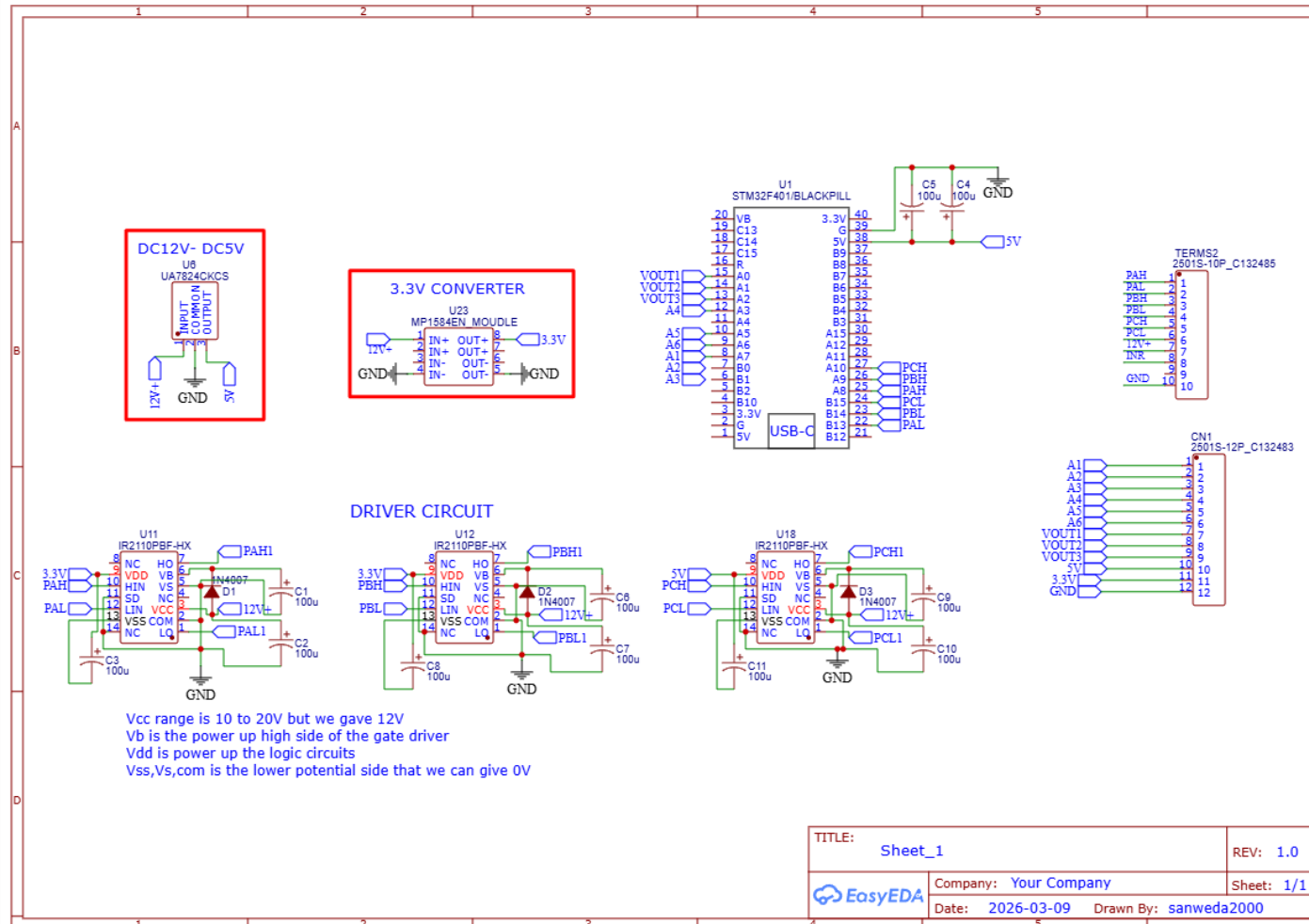


Figure 1: Schematic diagram of Controlling circuit

1.2 PCB design

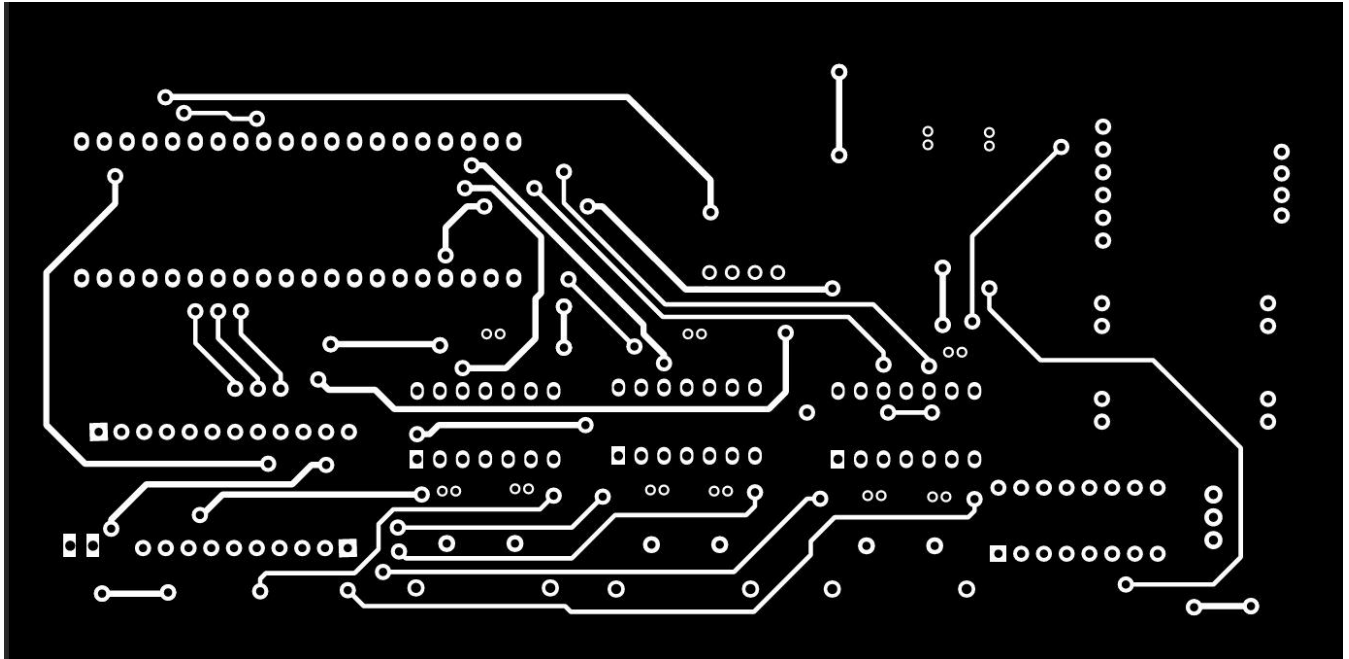


Figure 2: Controller circuit : PCB : Layer 1

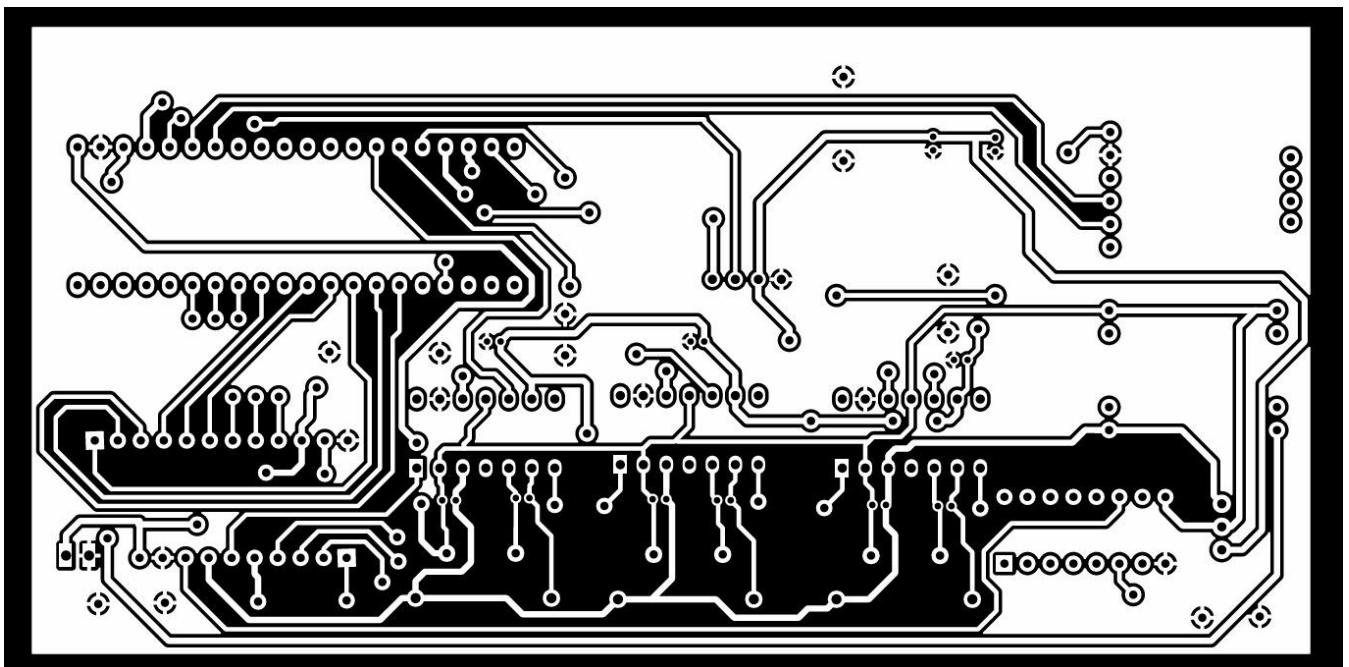


Figure 3: Controller circuit : PCB : Layer 2

1.3 Final PCB

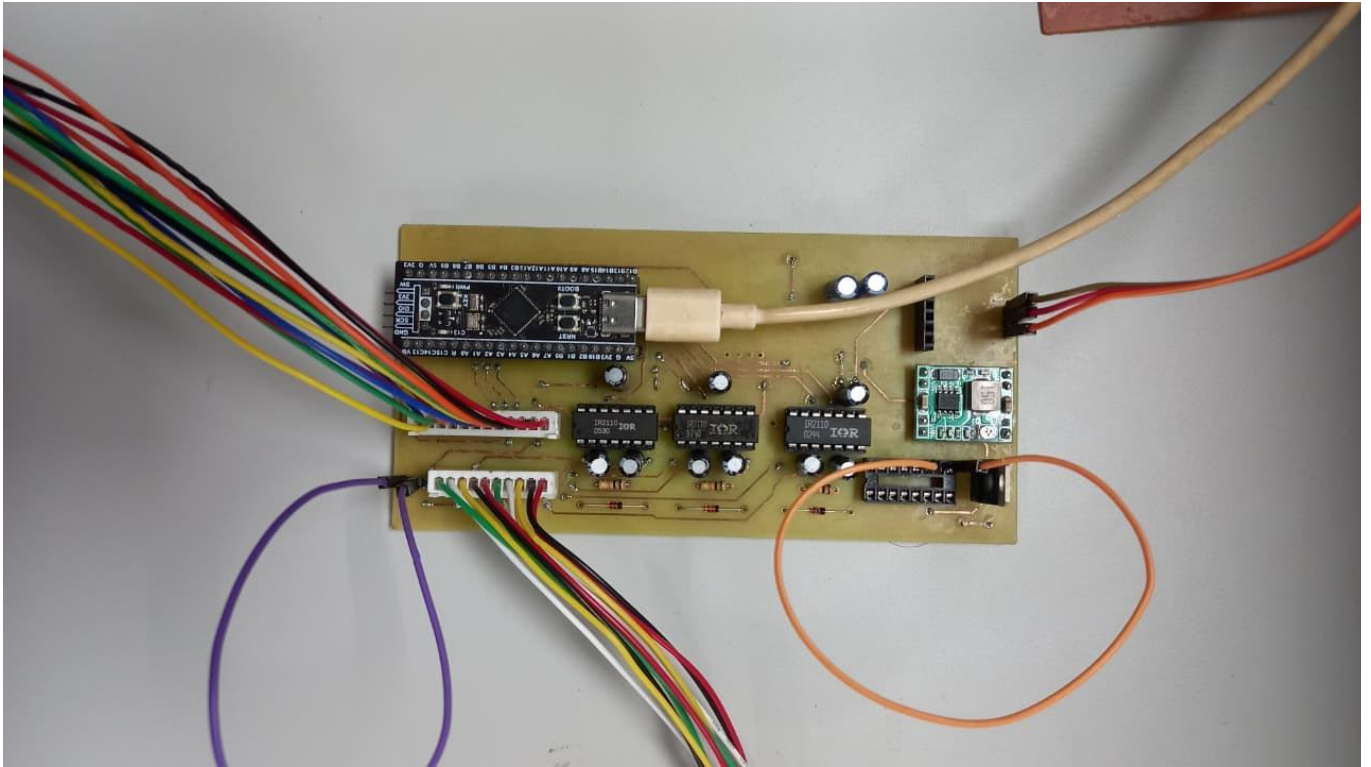


Figure 4: Hardware - Controller circuit

1.4 Operation

The PWM signals generated by the STM32 microcontroller have a logic-level voltage of 3.3 V, they do not have ability to directly drive MOSFET gates efficiently. Therefore, a gate driver IC (IR2110) is employed between the microcontroller and the MOSFETs to amplify the gate signals to the required voltage level (10-20 V for IRFP460 MOSFET). This ensures fast switching, reduced switching losses, and reliable operation of the inverter.

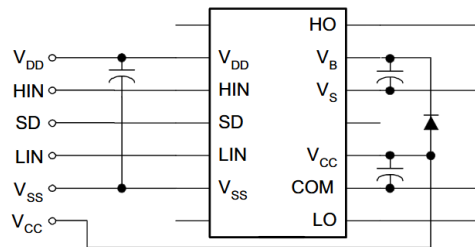


Figure 5: Gate driver circuit

- Voltage regulators

An MP1584EN DC–DC buck converter is used to step down 12 V supply to a regulated 3.3 V output, providing the operating voltage required by the IR2110 gate driver IC. (The inverter control circuit includes a 12 V auxiliary power supply)

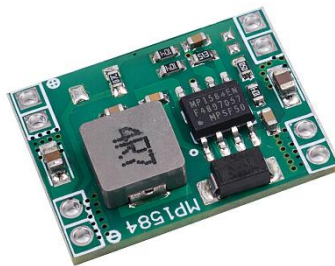


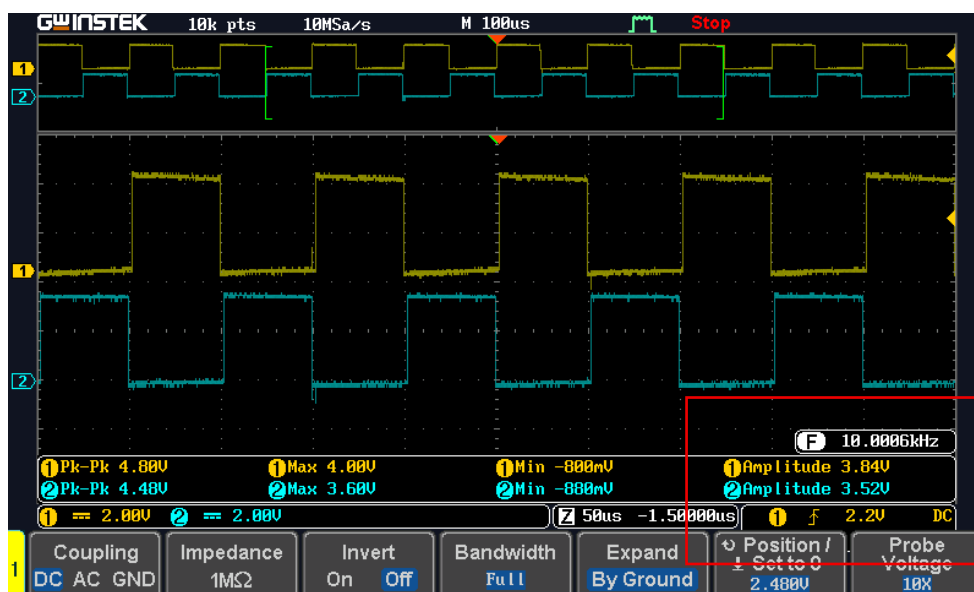
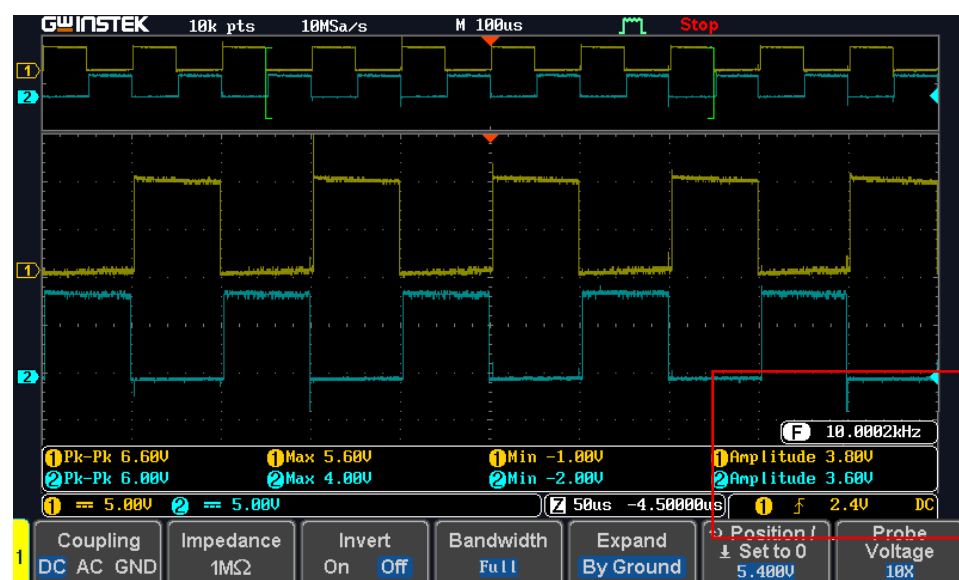
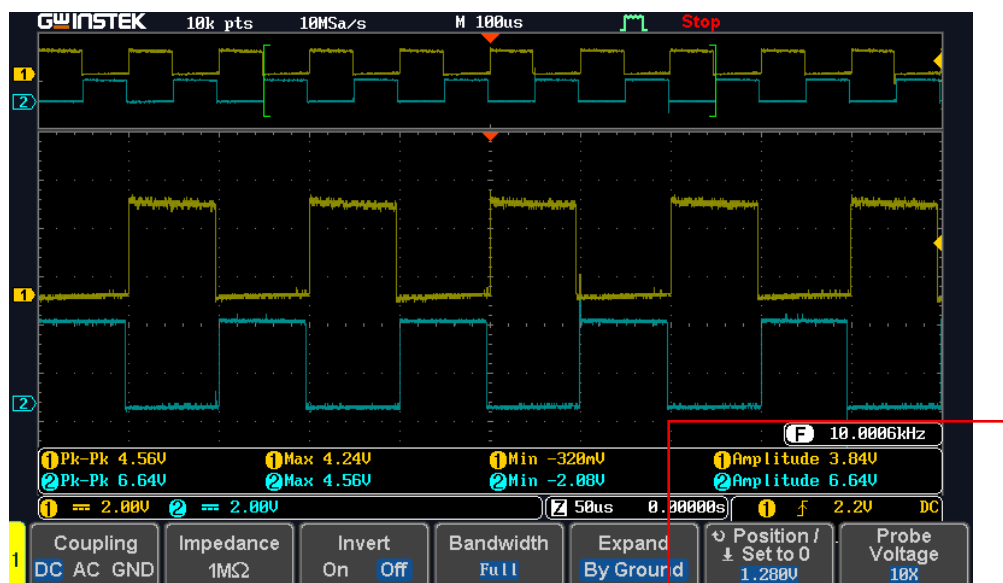
Figure 6: MP1584EN DC-DC buck converter

The voltage and current sensing circuits require a regulated 5 V DC supply for proper operation. This supply is provided by the UA7805CKCS linear voltage regulator, which steps down the available input voltage to a stable 5 V output. Since the regulator dissipates power in the form of heat during operation, a heat sink is attached to the UA7805CKCS to ensure effective thermal management, prevent overheating, and maintain reliable performance

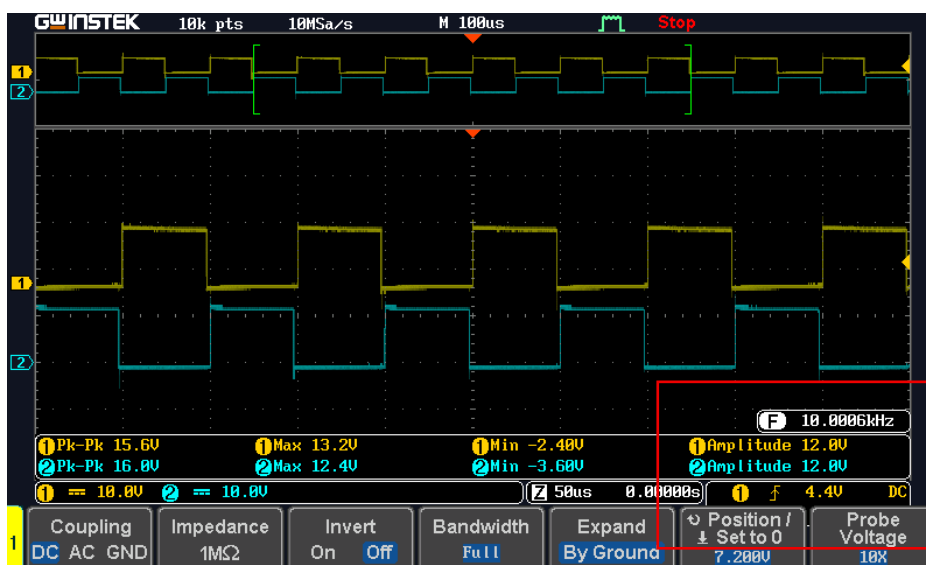
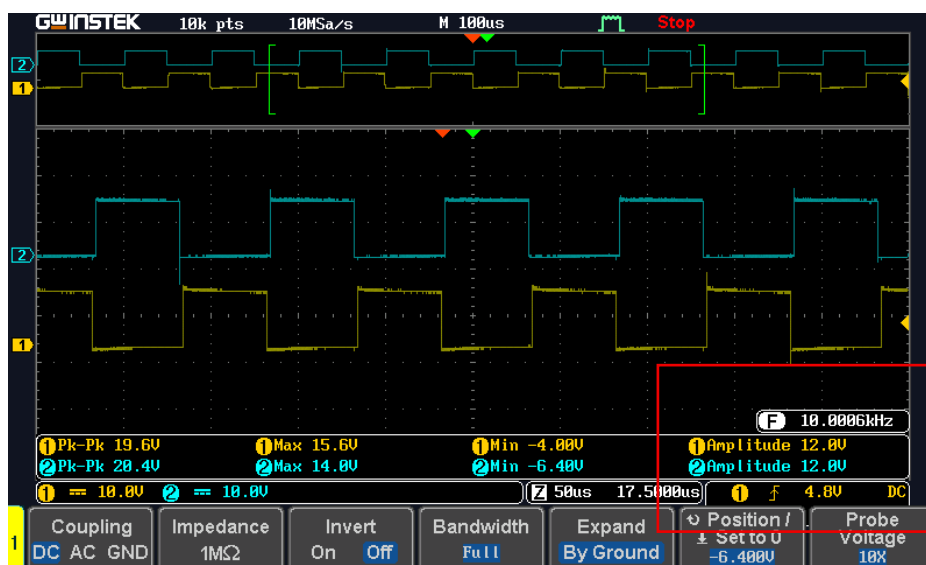
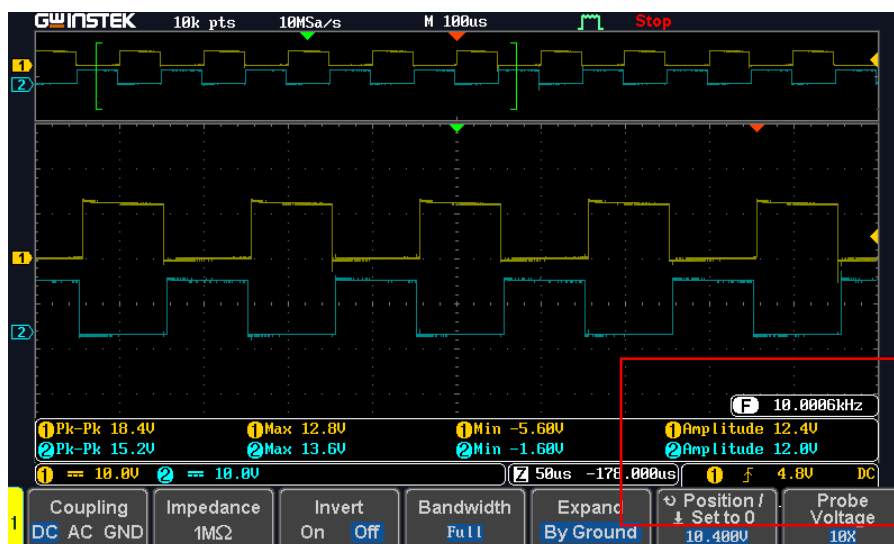


Figure 7: UA7805CKCS

PWM before gate driver circuit



PWM after gate driver circuit



1.5 Components

Table 1: Components of Controller Circuit

Component	Value/model name/parameters	Datasheet
Microcontroller	STM32F411	https://www.st.com/resource/en/datasheet/stm32f411ce.pdf
Gate driver IC	IR2110	https://www.alldatasheet.com/datasheet-pdf/view/82793/IRF/IR2110.html
Filter capacitor for stm32	100uF / 16V	-
Filter capacitor for IC	10uF/25V	-
Module to convert 12v DC-3.3v DC	MP1584EN	https://www.alldatasheet.com/datasheet-pdf/view/551593/MPS/MP1584EN.html
Module to convert 12v DC-5v DC	UA7805CKCS	https://www.alldatasheet.com/datasheet-pdf/view/230309/TI/UA7805CKTTR.html
Heat sink for UA7805CKCS	-	-

2.1 Schematic Diagram

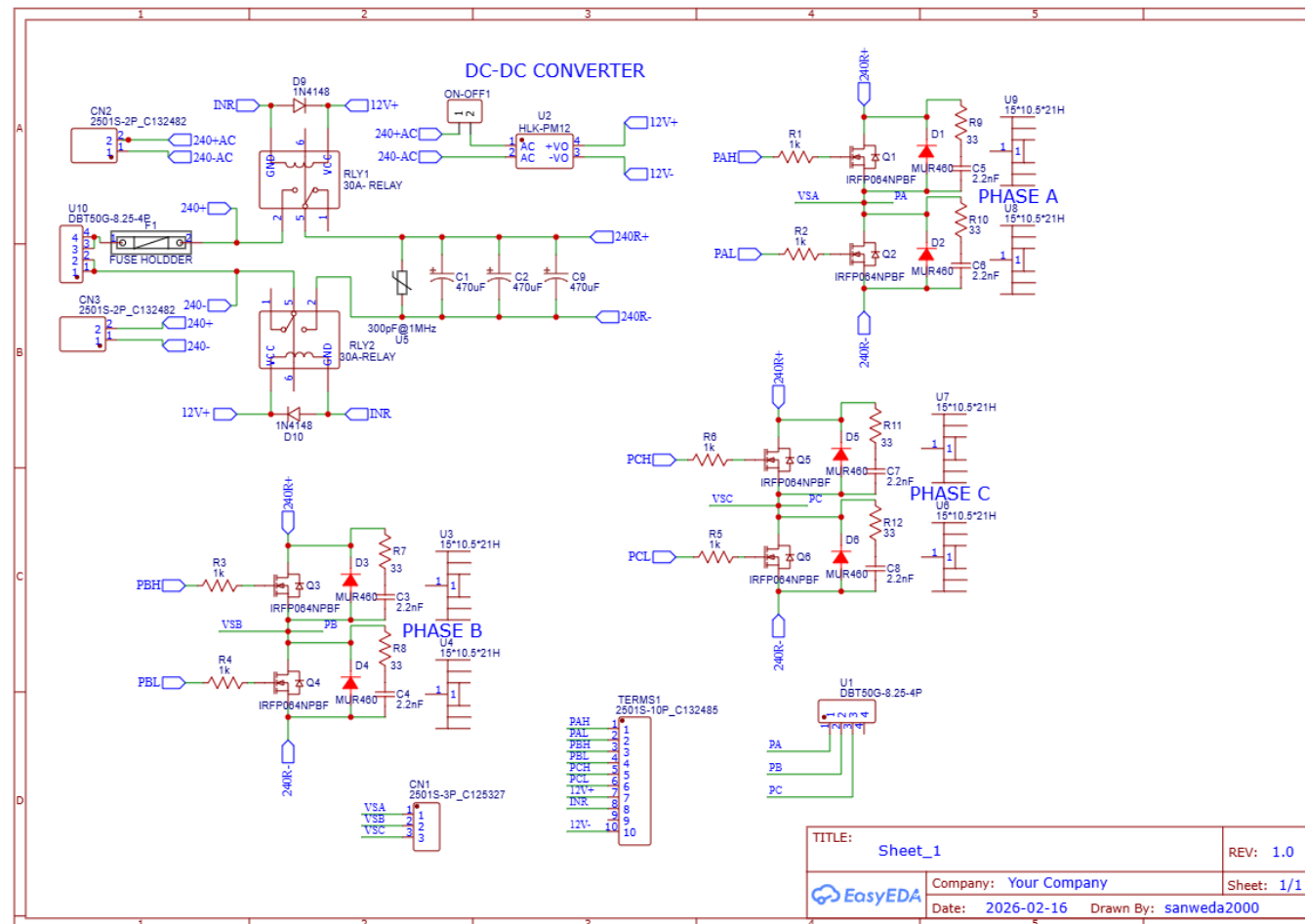


Figure 8: Schematic diagram of Inverter circuit

2.2 PCB design

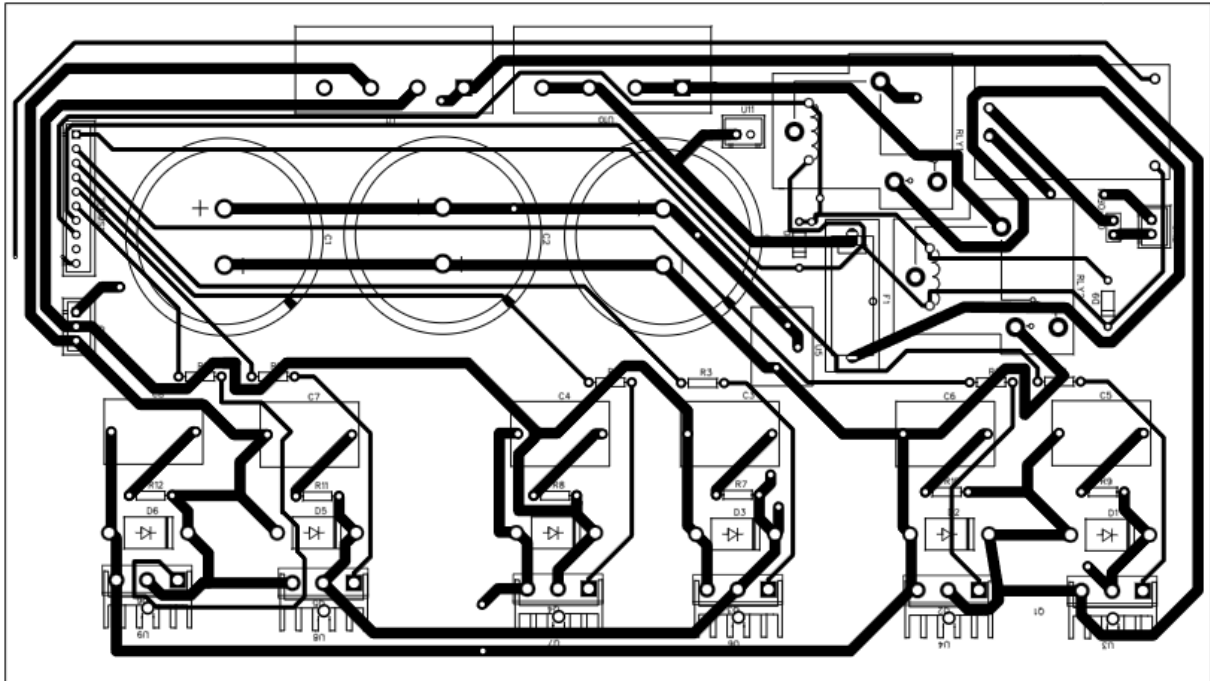


Figure 9: PCB design of Inverter circuit

2.3 Final PCB

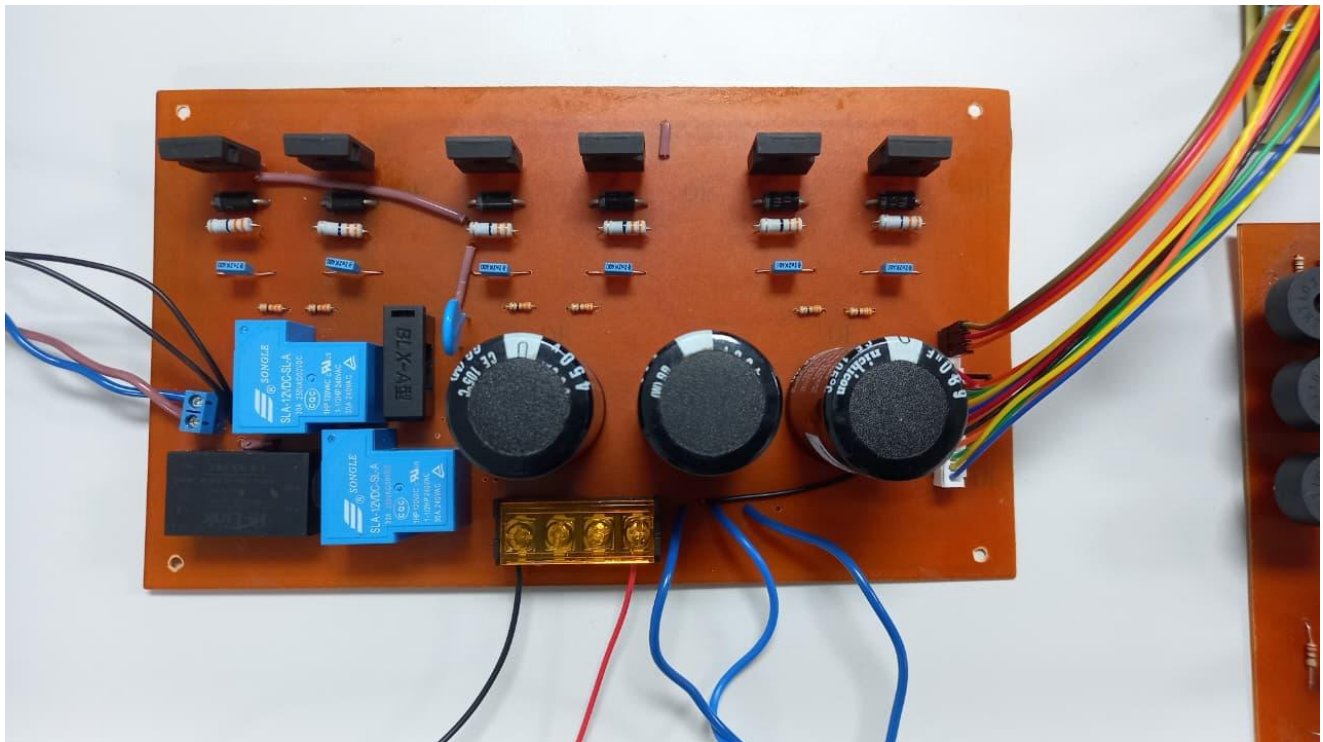


Figure 10: Hardware circuit - Inverter circuit

2.4 Operation

MOSFET Switching Operation

The inverter employs six MOSFETs configured as three half-bridge inverter legs. Each leg consists of one high-side MOSFET and one low-side MOSFET.

For each phase:

- The high-side and low-side MOSFETs operate complementarily.
- When the upper MOSFET is turned ON, the lower MOSFET is OFF.
- When the lower MOSFET is ON, the upper MOSFET is OFF.
- A dead time is inserted between switching transitions to prevent shoot-through, where both MOSFETs conduct simultaneously.

Relay Circuit

Three electromagnetic relays are incorporated into the inverter board to control the connection of the inverter output.

The relay circuit serves several purposes:

- Isolates the inverter during startup and shutdown,
- Prevents accidental energization of the output,
- Allows safe connection and disconnection of external loads,
- Protects the inverter during abnormal operating conditions.

During system initialization, the relays remain open until the controller confirms stable operation. Once the DC power supply reach the desired values, the relays are energized to connect the inverter output to the load. Similarly, under fault conditions or during shutdown, the relays disconnect the load to enhance system safety.

Snubber Circuit

Each MOSFET switching leg incorporates a snubber circuit consisting of a capacitor and associated components connected across the switching device.

The snubber circuit performs several important functions:

- Suppresses high-frequency voltage spikes generated during MOSFET switching,
- Limits the voltage overshoot caused by parasitic inductances,
- Reduces electromagnetic interference (EMI),
- Minimizes switching stress on the MOSFETs,
- Improves the overall reliability and lifetime of the inverter.

When a MOSFET switches rapidly, energy stored in stray inductances can produce transient overvoltage. The snubber capacitor absorbs this energy and reduces the rate of voltage change (dv/dt), thereby protecting the switching devices from excessive voltage stress.

DC-Link Capacitors

The large electrolytic capacitors connected across the DC bus act as the DC-link capacitor bank.

Their functions are to:

- Smooth the DC input voltage,
- Supply instantaneous current during switching,
- Reduce DC voltage ripple,
- Stabilize the DC bus voltage,
- Improve inverter performance during load transients.

These capacitors provide the energy buffer required between the DC source and the inverter bridge.

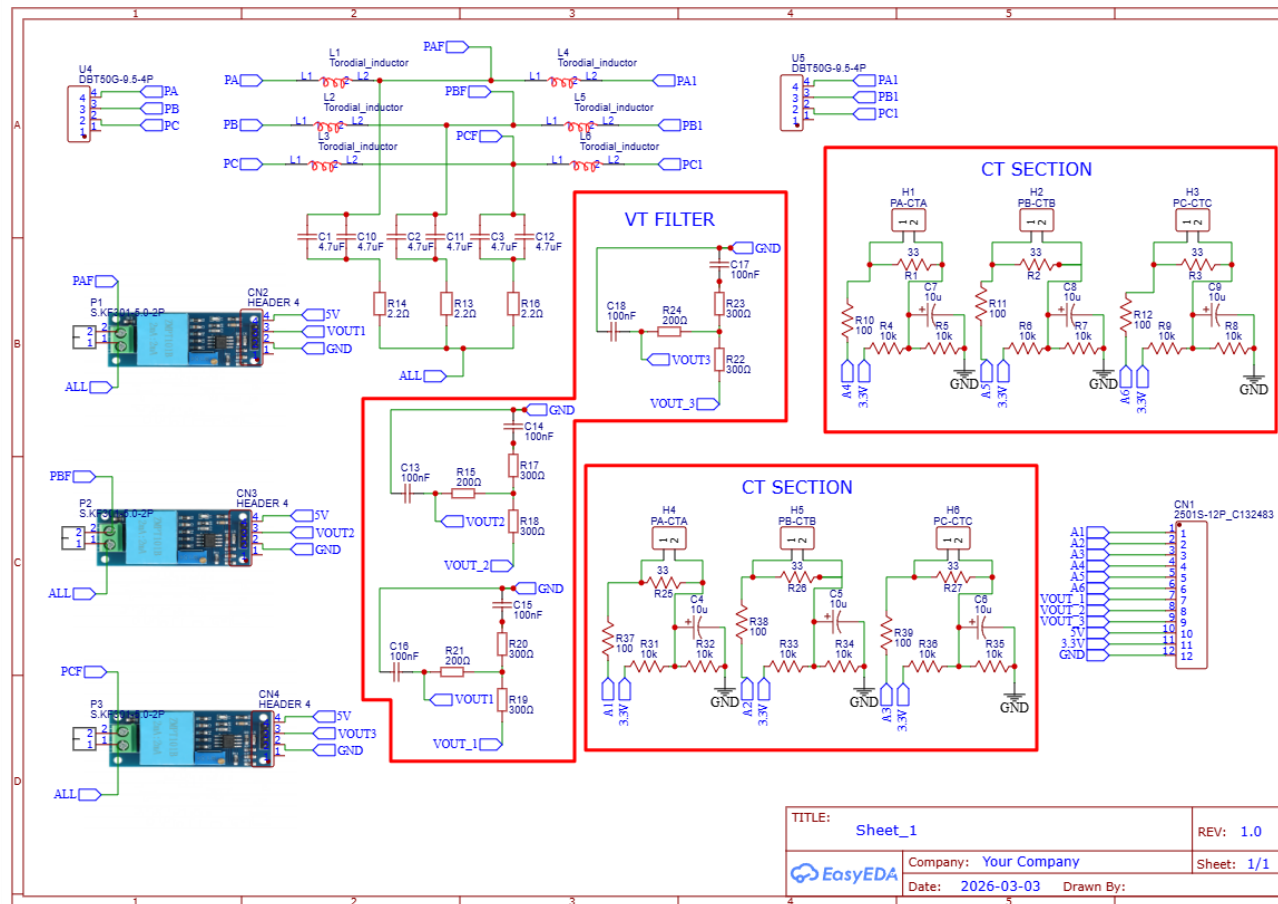
2.5 Components

Table 2: Components of Inverter Circuit

Component	Value	Data sheet
MOSFET	IRFP460	https://www.alldatasheet.com/datasheet-pdf/view/67546/INTERSIL/IRFP460.html
230 V AC – 12 V DC	HLKPM12	https://nettigo.pl/attachments/503
DC link capacitor	640 μ F / 450V	-
Relay	12 V DC / 30 A	-
Fuse	5A	-
Varistor	14D471K /470 V	https://www.alldatasheet.com/datasheet-pdf/view/555630/WPI/14D471K.html
Ultrafast recovery rectifier diode	MUR460	https://www.alldatasheet.com/datasheet-pdf/view/1397411/VISHAY/MUR460.html
Gate resistor	33 ohm , 1/4 W	-
Resistor Snubber	33 ohm , 2 W	-
Film capacitor Snubber	2N2K400 / 2.2 nF	-

3 LCL filter + Voltage and Current measure circuit

3.1 Schematic Diagram



3.4 Operation

- LCL filter circuit

The PWM output generated by the three-phase inverter contains the required 50 Hz fundamental component together with high-frequency switching harmonics produced by the 10 kHz switching operation of the MOSFETs. These harmonics must be attenuated before supplying power to the load.

The LCL filter, consisting of two inductors (L_1 and L_2) and a shunt capacitor (C_f) with damping resistor (R_f), performs this function. The inverter-side inductor (L_1) limits the high-frequency current ripple generated by the inverter switches, while the grid/load-side inductor (L_2) further suppresses the remaining harmonic components before they reach the load. The filter capacitor provides a low-impedance path for the high-frequency switching harmonics, diverting them away from the load. To prevent resonance within the LCL filter, a damping resistor is connected in series with the capacitor, ensuring stable operation and improving the transient response of the system.

As a result, the high-frequency PWM waveform produced by the inverter is converted into a balanced three-phase sinusoidal voltage suitable for supplying islanded loads.

- Voltage measuring circuit

The inverter output voltage is continuously monitored using three ZMPT101B voltage sensor modules, one for each phase. These sensors provide galvanic isolation between the high-voltage power circuit and the low-voltage control circuit, ensuring safe operation of the microcontroller.

Each ZMPT101B module employs a precision voltage transformer to step down the AC voltage and generate a low-voltage analog signal proportional to the measured phase voltage. The conditioned output signal is then sampled by the Analog-to-Digital Converter (ADC) channels of the STM32F411 microcontroller.

The measured phase voltages are used as feedback signals in the voltage control loop, where they are transformed into the synchronous dq reference frame for closed-loop voltage regulation. In addition, the measured voltages are transmitted and used to calculate the final voltage to the grid to the monitoring dashboard for real time visualization.

- Current measuring circuit

The inverter output currents are measured using three ZMCT103C current transformer (CT) modules, one for each phase. These sensors provide non-invasive current measurement while maintaining complete electrical isolation between the power stage and the control electronics.

Each current transformer converts the measured AC phase current into a proportional low-level current signal. This signal is converted into a measurable voltage using a burden resistor and conditioned before being sampled by the ADC channels of the STM32F411 microcontroller.

The measured current signals are transformed into the synchronous dq reference frame and utilized by the inner current control loop to regulate the inverter-side currents. They are also used for system monitoring, performance evaluation, and protection against abnormal operating conditions such as overcurrent.

3.5 Components

Table 3: Components of LCL + Voltage and current measure circuit

Component	Value	Datasheet
Inductors	2.3 mH	-
Film capacitor	4.7 uF /400V	-
Ceramic resistor	2.2 ohm /10 W	-
Current transformer	ZMCT103C / 5A	https://www.alldatasheet.com/datasheet-pdf/pdf/1821417/MICRO-TRANSFORMER/ZMCT103C.html
Voltage sensor module	ZMPT101B	https://www.alldatasheet.com/datasheet-pdf/view/1821973/MICRO-TRANSFORMER/ZMPT101B.html
Capacitor	10uF /25 V	-
Ceramic capacitor	0.1 uF	-
Resistor	33 ohm, 1/4 W	-
Resistor	100 ohm , 1/4 W	-
Resistor	10k	-
Resistor	1k	-